

Hydrodynamic Characterization of the FF11K Marine Lander

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MIT-Japan 2025

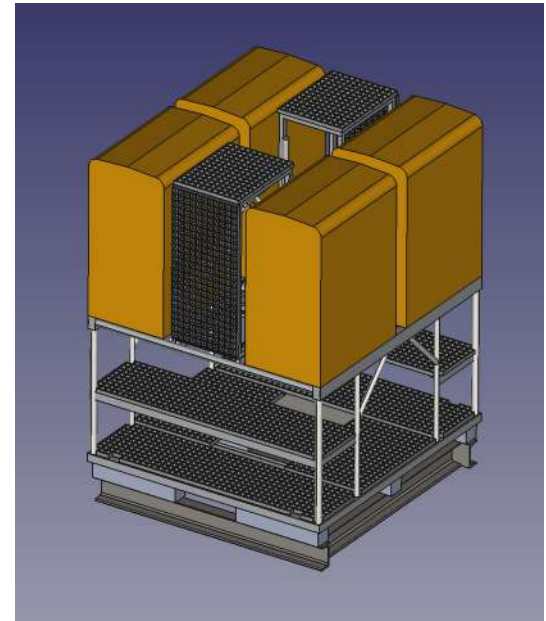
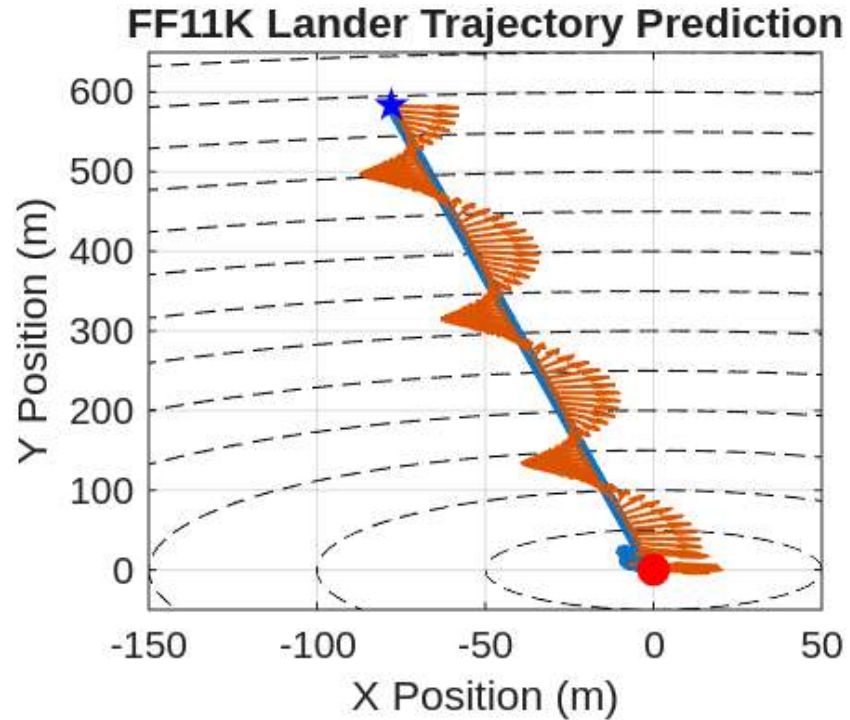


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Introduction

- The **FF11K** is a **F**ree **F**alling Lander developed by JAMSTEC [1]
- Drops to **11 Kilometer** depths to study the Hadal Zone
- Can be equipped with new sensors each mission
- Sinks from surface and drops ballast to rise again



Figure 1:
FF11K Lander [1]

Introduction

- The **FF11K** is a **Free Falling** Lander developed by JAMSTEC [1]
- Drops to **11 Kilometer** depths to study the Hadal Zone
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Size [m]	1.6×1.7×2.2
Dry weight [kg]	2000 (with ballast), 1700 (w/o ballast)
Weight in surface water [kg]	110 (with ballast), -145 (without ballast)
Maximum depth [m]	11000
Standard equipment	Teledyne Benthos R12K and 865A
	SBE49 CTD sensor
	HD camera (Sony EB6500)
	200W LED light ×2
	Iridium beacon
	Flasher
	Acoustic modem

Figure 2:
FF11K Specifications [2]

Introduction (Continued)

- Lander drifts significantly when dropped in water
- Main drift occurs upto **1.5 km** depths, where ocean currents are strongest
- Accurate model of lander's movements needed for missions

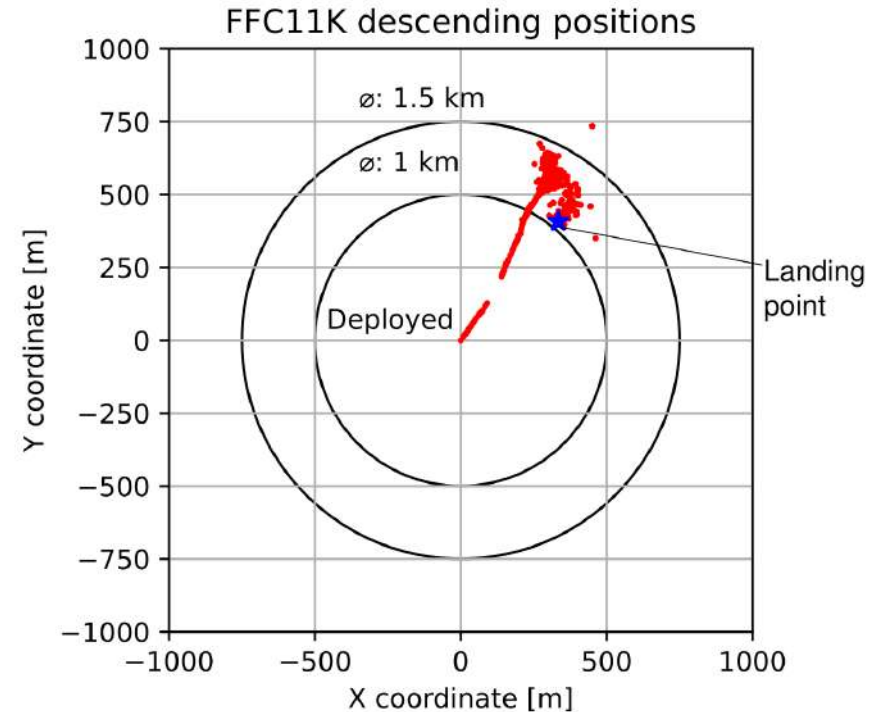
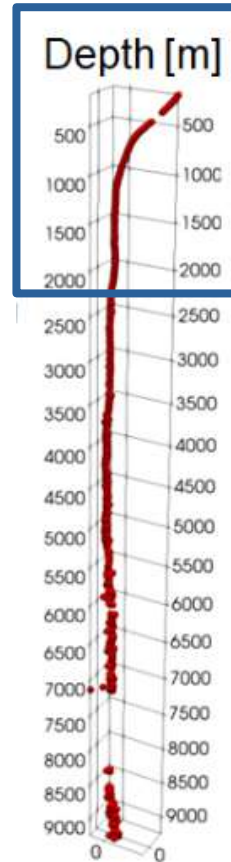


Figure 3:
FFC11K Lander's Descending Trajectory
(3D and Top View) [1]

Introduction (Continued)

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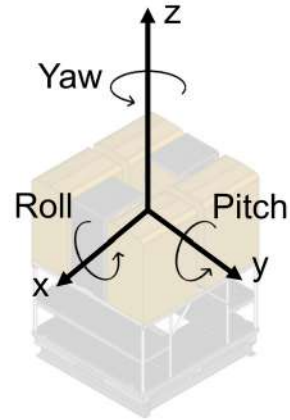
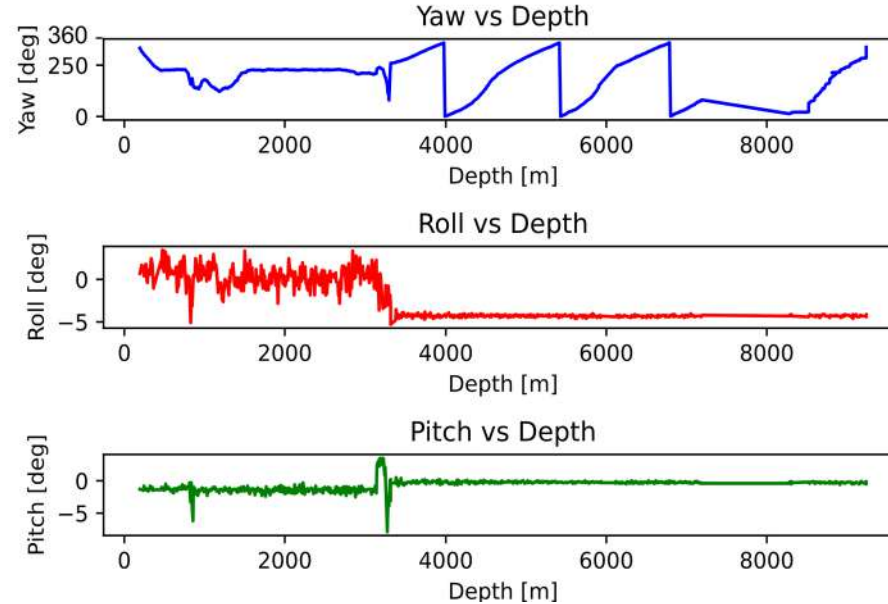


Figure 4:
FF11K Lander's RPY Angles during Descent [1]

Lander Physics

$$(M_{rb} + M_a)\dot{\mathbf{v}} + C(\mathbf{v})\mathbf{v} + \mathbf{D}(\mathbf{v})\mathbf{v} + \mathbf{g}(\mathbf{n}) = \boldsymbol{\tau} \quad [3]$$



Hydrodynamic Mass (M):	Inertia effect from lander and fluid	(Y)
Coriolis Matrix (C):	Assumed small (Ignore for now)	(N)
Drag Matrix (D):	Resistance from surrounding fluid	(Y)
Restoring Forces (g):	Gravity & Buoyancy (keep lander vertical)	(Y)
Input Forces (τ):	Control System Input (0 for now)	(N)



$$(M_{rb} + M_a)\dot{\mathbf{v}} + \mathbf{D}(\mathbf{v})\mathbf{v} + \mathbf{g}(\mathbf{n}) = \mathbf{0}$$

Lander Physics (Continued)

- Observations from previous tests show that Roll and Pitch ≈ 0 (4 DoF system)
- Coupled Hydrodynamics were not considered due to time constraints
- **Main goal:** Study Forces and Torques acting on lander during descent
- **CFD** is a good method for research with given time-frame and budget

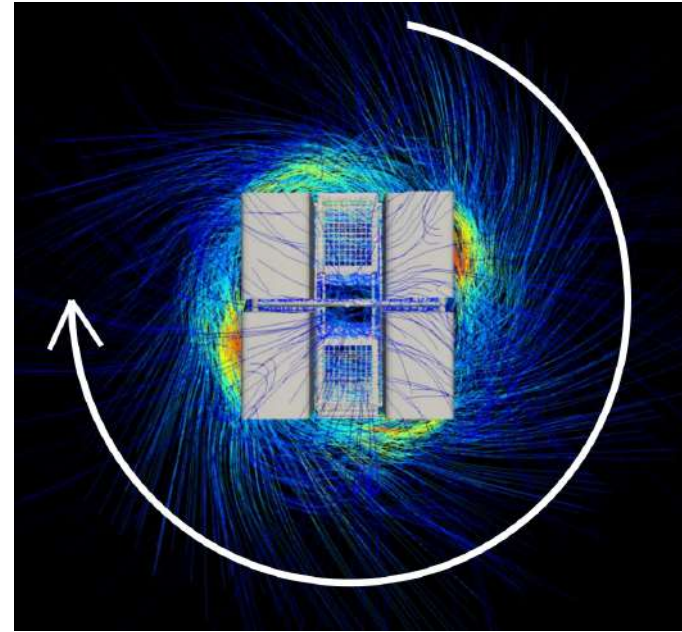


Figure 5:
Lander can translate in XYZ and rotate around Z axis during descent [1]

OpenFoam

- **Open Field Operation And Manipulation** (OpenFOAM) is a set of tools for solving complex fluid dynamics problems [4]
- SimpleFoam (steady-state) & PimpleFoam (transient)
- κ - ω SST Reynolds Averaged Navier Stokes
- Solves large system of Navier Stokes Equations (Partial ODE's) to obtain pressures (**forces**), fluid speeds, etc.
- **Detail vs. Time**

Open  FOAM [5]

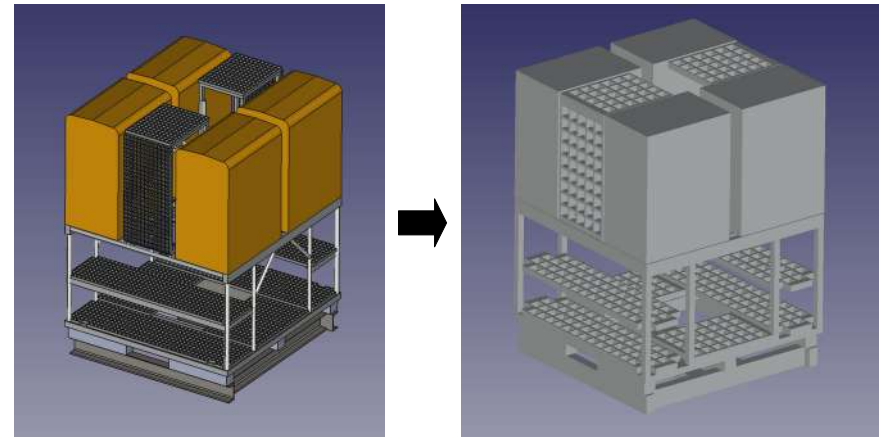


Figure 6:
Simplified Models of FF11K Lander
X: 1.590 m, Y: 1.660 m, Z: 2.2 m

Methods

Drag Forces

- **SimpleFoam** is a commonly-used solver for finding steady-state forces
- Fit data to simplified drag equation
- **Attempt #1:** Fluid speed set to 0.25 m/s in XYZ Directions
 - Needed more data and set model for next simulations
- **Attempt #2:** Wrote Bash script to automatically run cases in all directions for speeds = [0.2 - 1]

$$F_d = \frac{1}{2} \rho_w v^2 C_d A_r \quad [6]$$



$$F_d = k_1 v + k_2 |v|v$$

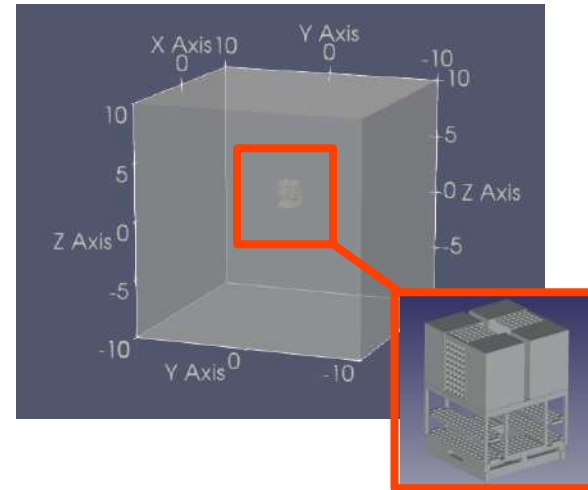


Figure 7:
Simplified Lander
inside bounding box

Drag Torque

- Two methods for running simulation:
 - Rotate fluid:
Multiple Reference Frame (MRF)
 - **Rotate lander:**
Arbitrary Mesh Interface (AMI)
- **Attempt #1:** Rotated **lander** at velocities [2 – 10] rad/s

$$F_d = k_1 v + k_2 |v|v$$



Figure 8:
Lander inside AMI boundary

Added Mass

$$M = \frac{F_{sim} - F_d}{a_{sim}} - \rho V \quad [7]$$

- **PimpleFOAM** is a commonly-used transient solver (speed and forces change over time)
- **Attempt #1:** Accelerated **lander** for 2 seconds each XYZ direction
 - Point Displacement Map (vs. Dynamic Mesh)
 - Very computationally expensive
 - Did not use proper acceleration profile [8]
- **Attempt #2:** Experimented with different **fluid** accelerations and directions
 - Pressure values diverged

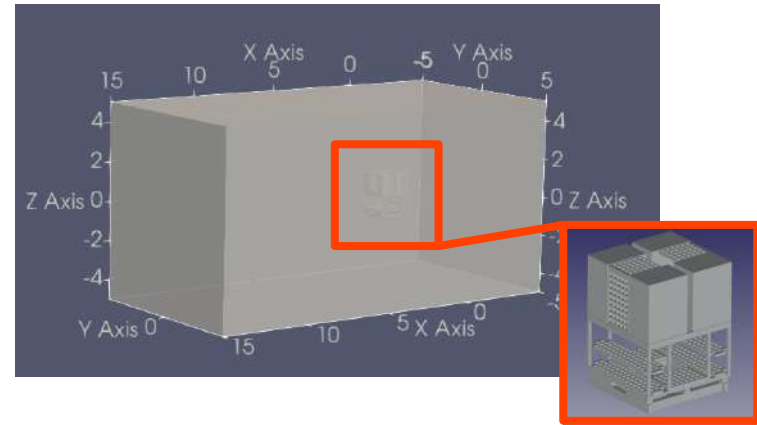
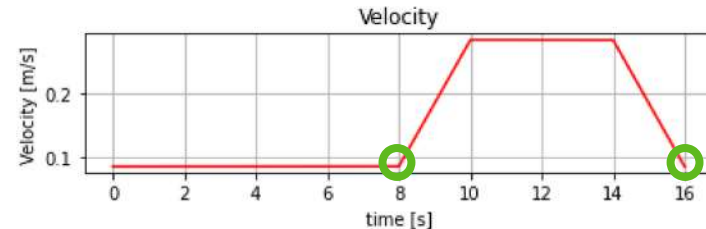


Figure 9:
Lander inside extended bounding box

Figure 10:
Attempt #2 Current Speed profile [8]



Current-Induced Torque

- Currents misaligned with lander's main axes may be responsible for some large magnitude torques
- Due to the lander's complex geometry, deriving a mathematical model of the torque is very difficult
- **Attempt #1:** Automatically ran 40 cases at different speeds and angles

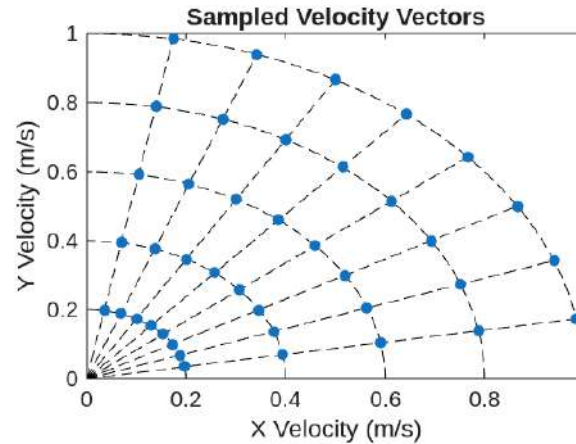


Figure 11:
Velocity vectors sampled for studying external torque

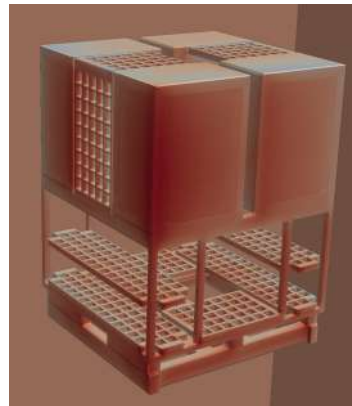


Figure 12:
Lander shaded based on pressure values. When integrated, these evaluate to forces and torques

Results

Drag Forces

Force Direction (XYZ, Yaw) →

- Quadratic equation was a good fit for diagonal entries
- Non-diagonal entries are small and flip signs sometimes
- Non-diagonal entries assumed negligible for modeling

Current Direction (XYZ)

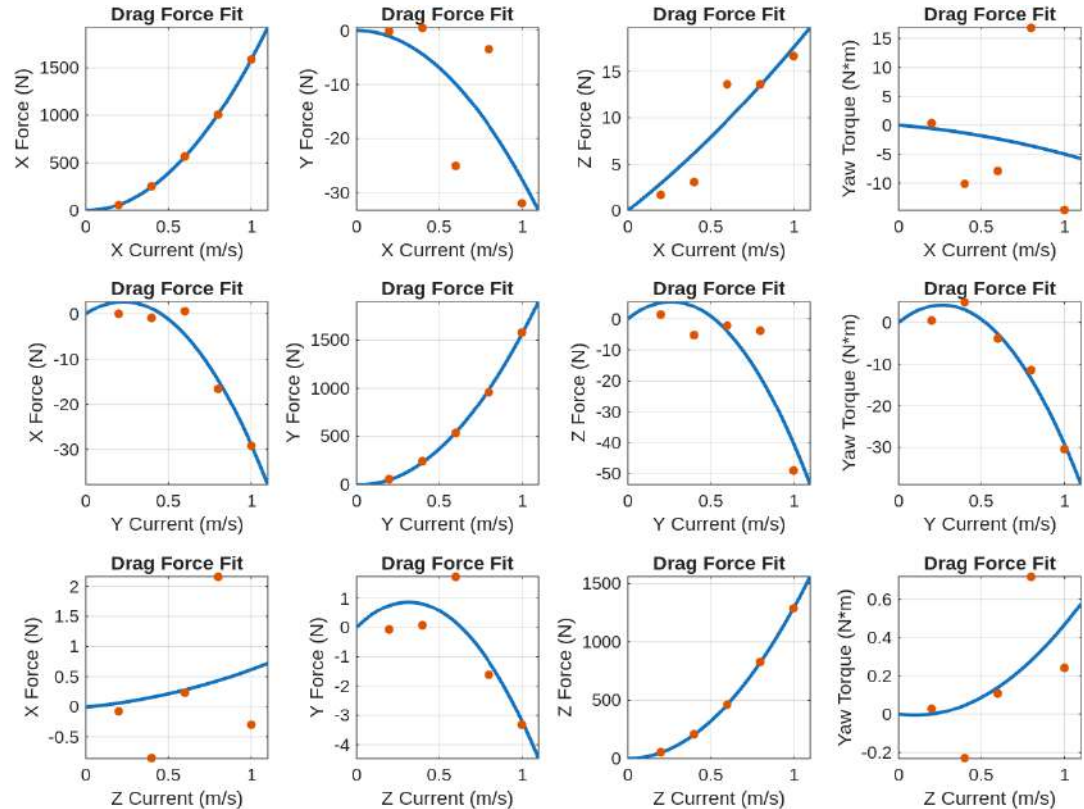


Figure 13:

Current speed vs. Drag forces and torques on lander

Drag Torques

- **Reminder:** This is **damping torque** resisting rotation of the lander.
- Large magnitude torques resist the lander's rotation.
- Same fit works for low and high speed rotations.

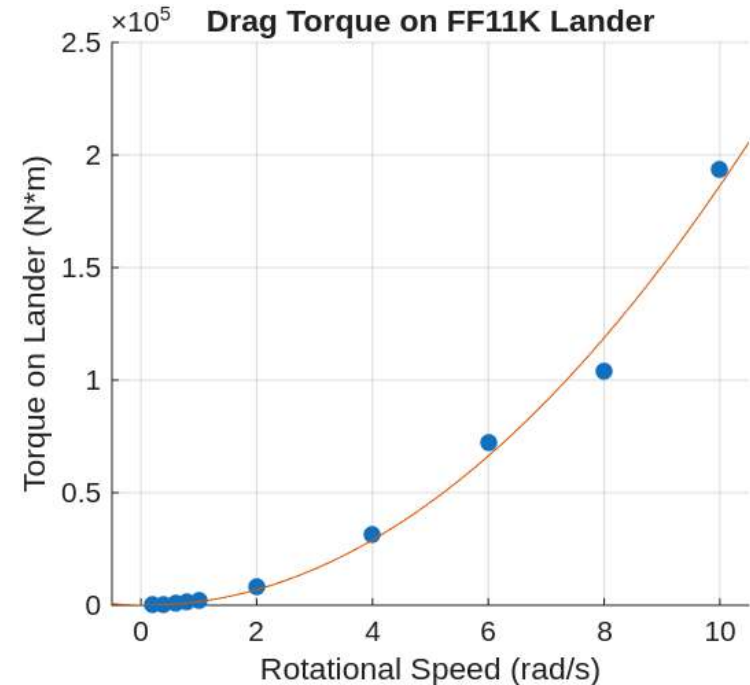
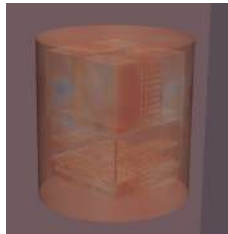


Figure 14:
Torque due to rotational damping on lander

Current-Induced Torque

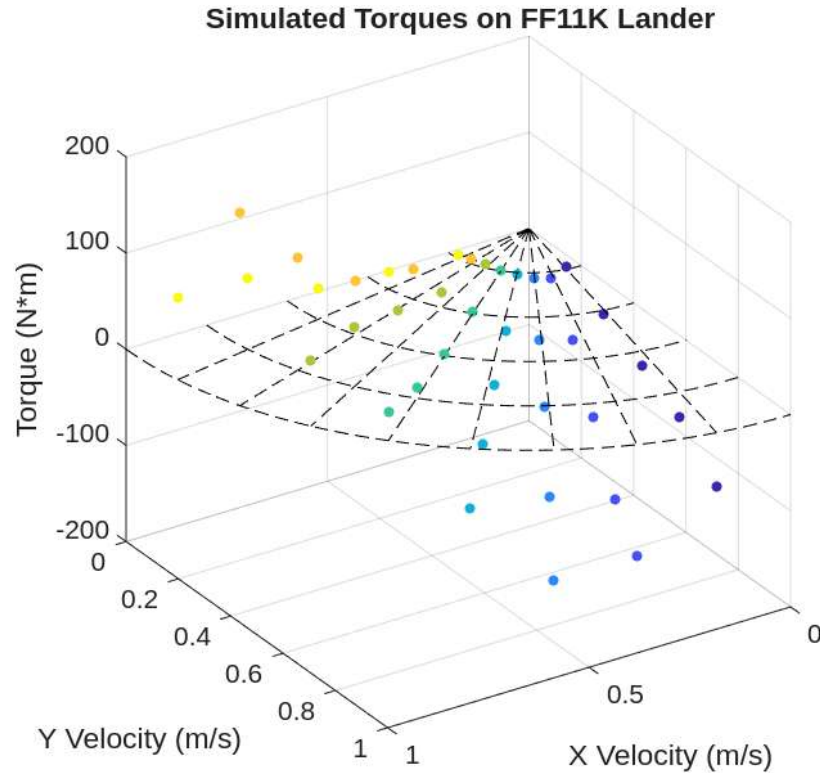


Figure 15:
Simulated torques on FF11K Lander

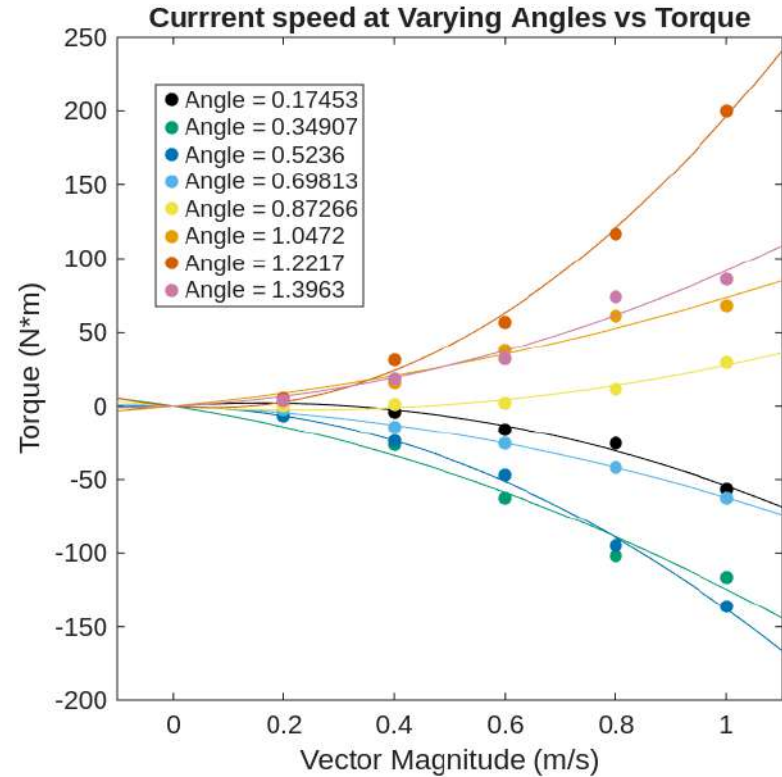


Figure 16:
Quadratic fit for vector magnitude vs. torque

Current-Induced Torque

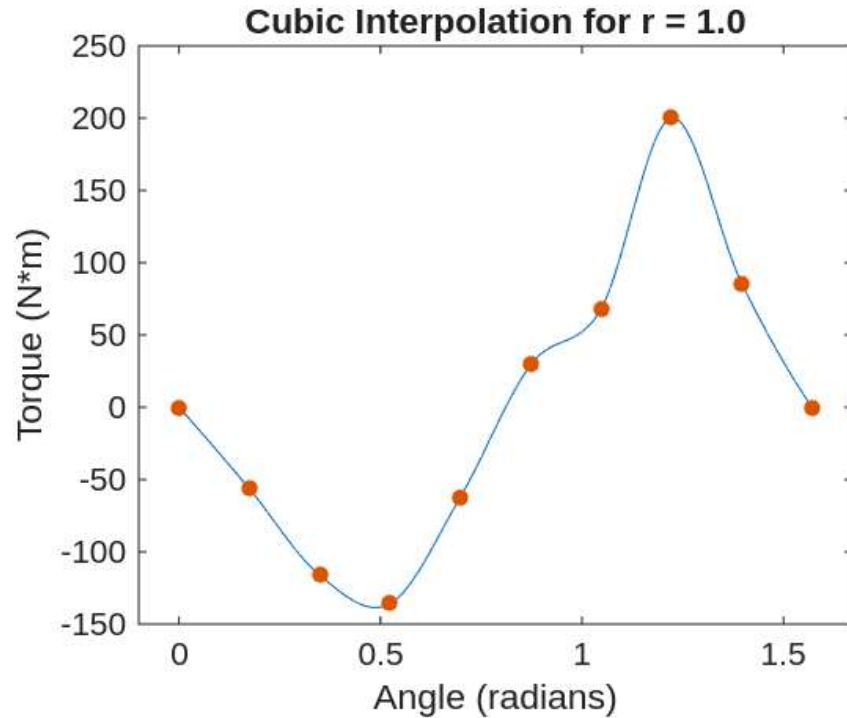


Figure 17:
Cubic Interpolation of torque samples from previously shown quadratic fits.

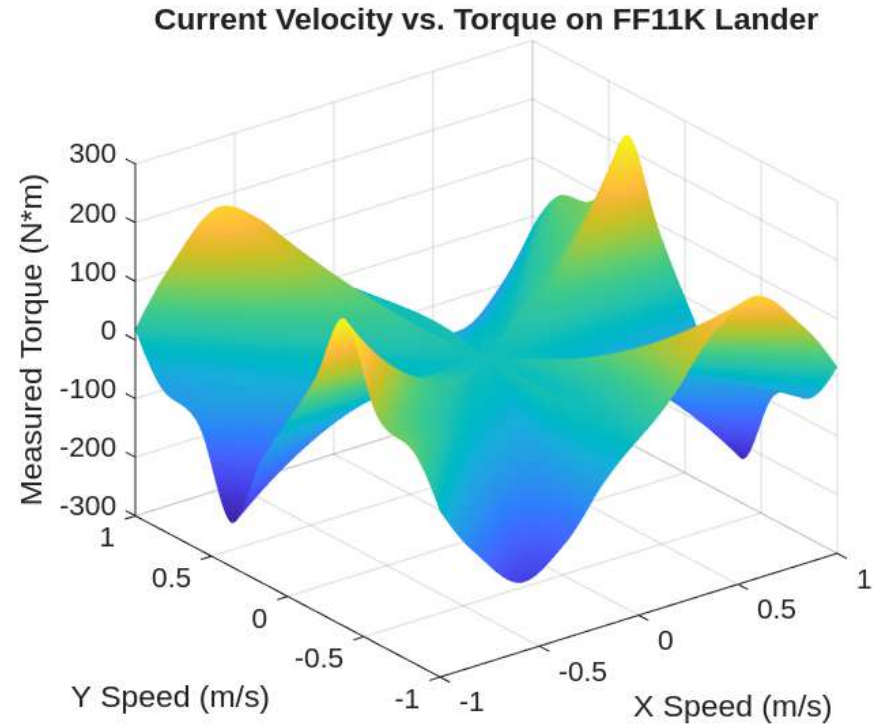


Figure 18:
Surface resulting from using quadratic fit on cross sections and interpolating along magnitude

Added Mass

- Attempt #1:
 - Accelerated lander (pointDisplacement)
 - No literature supporting this method
- Attempt #2:
 - Pressures > 1e+23

```
M = 4x4
104 x
      0.1789      0      0      0
      0      0.1383      0      0
      0      0      0.1431      0
      0      0      0      1.4110
```

Figure 19:
Hydrodynamic Mass matrix calculated from
pointDisplacement Method

Trajectory Prediction

- Used current speed and direction data from Yokosuka 25-12
- Euler's & Heun's methods of integration give similar results
- Largest Drift after 1500m

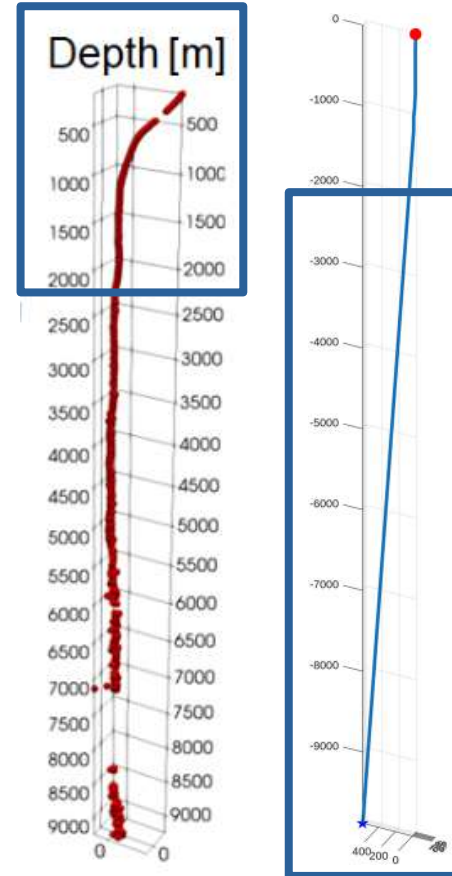


Figure 20:
3D View of Measured
Trajectory vs. Predicted
Trajectory
(Different currents)

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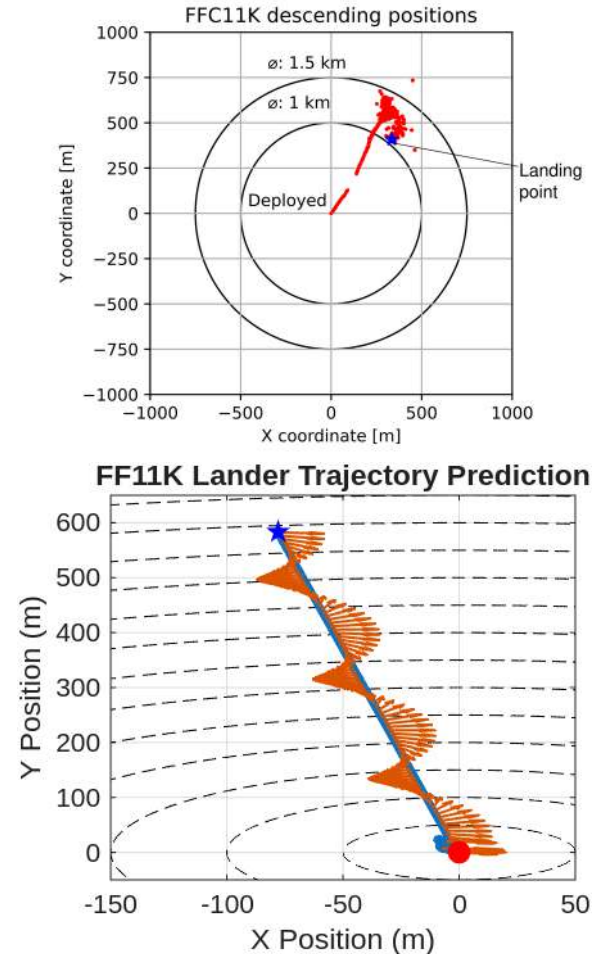
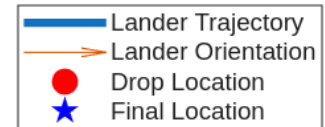
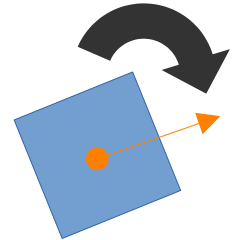


Figure 21:
Top View of
Measured
Trajectory vs.
Predicted
Trajectory
(Different
currents)



Summary

- Good results for Drag matrix and external torques
- Set up a framework for automatically running simulations
- Inertial terms in system of equations remain to be studied, but cases were set up
- Trajectory prediction needs further improvements
- Future work...

Results

- Drag Matrix
- Current-Induced T.
- Added Mass
- Trajectory Prediction

Yokosuka Cruise

- 10-Day Cruise to test Urashima AUV
- First time on a research vessel
- Very rewarding experience



その他 ...

金閣寺



箱根



富士山



大阪城



気仙沼



Bibliography

- [1] E. di Maria, T. Nakatani, Y. Maeda and T. Inoue, "Hydrodynamic Investigation of a Free Fall Lander for the Descent Trajectory Control," *OCEANS 2024 - Singapore*, Singapore, 2024, pp. 1-6, doi: 10.1109/OCEANS51537.2024.10682155.
- [2] Maeda, Y., Kida, Y., & Deguchi, M. (2020). Hadal free-fall camera lander transmitting hadal images via high-rate acoustic communication. *Sea Technology*, 61 (12), 22-24.
- [3] Fossen, T. I. (1999). *Guidance and Control of Ocean Vehicles*. John Wiley & Sons, Inc.
- [4] Weller, H. G., Tabor, G., Jasak, H., & Fureby, C. (1998). A tensorial approach to computational continuum mechanics using object-oriented techniques. *Computers in Physics*, 12(6), 620–631. <https://doi.org/10.1063/1.168744>
- [5] Aboukhadijeh, F. (2004). OpenFOAM. <https://www.openfoam.com>
- [6] Benson, T. (1999). The drag equation. NASA. <https://www.grc.nasa.gov/www/k-12/VirtualAero/BottleRocket/airplane/drageq.html>
- [7] F. H. Imlay, "The complete expressions for Added Mass of a rigid body moving in an ideal fluid". Report 1528, David Taylor Model Basin (DTMB), 1961.
- [8] E. Javanmard, S. Mansoorzadeh, J. A. Mehr. "A new CFD method for the determination of translational added mass coefficients of an underwater vehicle". *Ocean Engineering*, 2020.